

Habitual Chocolate Consumption May Increase Body Weight in a Dose-Response Manner

Objective Habitual chocolate intake was recently found to be associated with lower body weight in three cross-sectional epidemiological studies. Our objective was to assess whether these cross-sectional results hold up in a more rigorous prospective analysis. Methods We used data from the Atherosclerosis Risk in Communities cohort. Usual dietary intake was assessed by questionnaire at baseline (1987–98), and after six years. Participants reported usual chocolate intake as the frequency of eating a 1-oz (~28 g) serving. Body weight and height were measured at the two visits. Missing data were replaced by multiple imputation. Linear mixed-effects models were used to evaluate cross-sectional and prospective associations between chocolate intake and adiposity. Results Data were from 15,732 and 12,830 participants at the first and second visit, respectively. More frequent chocolate consumption was associated with a significantly greater prospective weight gain over time, in a dose-response manner. For instance, compared to participants who ate a chocolate serving less often than monthly, those who ate it 1–4 times a month and at least weekly experienced an increase in Body Mass Index (kg/m²) of 0.26 (95% CI 0.08, 0.44) and 0.39 (0.23, 0.55), respectively, during the six-year study period. In cross-sectional analyses the frequency of chocolate consumption was inversely associated with body weight. This inverse association was attenuated after excluding participants with preexisting obesity-related illness. Compared to participants without such illness, those with it had higher BMI and reported less frequent chocolate intake, lower caloric intake, and diets richer in fruits and vegetables. They tended to make these dietary changes after becoming ill. Conclusions Our prospective analysis found that a chocolate habit was associated with long-term weight gain, in a dose-response manner. Our cross-sectional finding that chocolate intake was associated with lower body weight did not apply to participants without preexisting serious illness.